

Here is how the CPU, RAM, and GPU work together to run battle royale games like PUBG or Free Fire:

CPU (Central Processing Unit)

The CPU acts as the brain of the operation, calculating the core game logic, player movements, and physics like bullet drop in real-time. It constantly processes your inputs and feeds instructions to the graphics card about what needs to be drawn next. Without a capable CPU, the game world wouldn't function, and the other components wouldn't know what to do.

RAM (Random Access Memory)

Think of RAM as your PC's high-speed workbench that temporarily holds all the active game data, like the map layout, textures, and player models. This allows the CPU and GPU to instantly access crucial assets without having to pull them from your slower permanent storage drive. Having enough RAM prevents the game from stuttering or freezing when you enter a new area or get into a heavily populated firefight.

GPU (Graphics Processing Unit)

The GPU is the heavy lifter for visuals, taking the raw instructions from the CPU and translating them into the detailed 3D world, lighting, and character models you actually see. It works rapidly to render these visual frames dozens or hundreds of times per second to ensure smooth gameplay. The more powerful your GPU, the higher the resolution and graphical settings you can push while maintaining a high frame rate.